

NavrasaNet: Emotional State Recognition via the Navarasa Doctrine

Shivansh Singh¹, Tharun Kumar Reddy², and Arihant Jain³

¹ Indian Institute of Technology Roorkee, Roorkee - 247667, INDIA
shivansh_s@ma.iitr.ac.in

² Indian Institute of Technology Roorkee, Roorkee - 247667, INDIA
tharun.reddy@ece.iitr.ac.in

³ Indian Institute of Technology Roorkee, Roorkee - 247667, INDIA
arihant_j@iks.iitr.ac.in

Abstract. Prevailing emotion recognition systems operate within the categorical boundaries inherited from Western psychological models, often unable to account for experiential states central to Indian aesthetic traditions. This paper introduces **NavrasaNet**, a deep learning system designed for real time recognition of emotional states organised according to the *Navarasa* theory of Bharata Muni's *Natya Shastra*. Built upon the **Navarasa Facial Expression Corpus (NFEC)**, a purpose built dataset developed in collaboration with classically trained Indian dance practitioners, thus ensuring both cultural and technical credibility. A critical portion of the dataset's foundational ground truth was provided by ICCR empanelled artist Smt. Anita Babu, bringing immense expressive authenticity to the corpus. Furthermore, to strengthen credibility of the dataset, it was verified by the government of India recognised scholar and researcher, **Dr. Manorama B.N.**. To capture subtle facial geometries while avoiding overfitting, we employ an EfficientNet B2 architecture utilizing a two phase fine tuning strategy with differential learning rates, augmented by label smoothing and dropout. Evaluating the model under a Leave One Subject Out (LOSO) cross validation scheme to ensure generalization to unseen performers, NavrasaNet achieves a peak validation accuracy of 93.43% and a robust test accuracy of 87.32% (with a macro averaged F1 score of 87%). Furthermore, the framework demonstrates practical utility through a Vedic Raga Recommender application.

Keywords: Emotion Recognition · Navarasa · EfficientNet B2 · Indian Knowledge Systems · Deep Learning · Affective Computing.

1 Introduction

Emotion recognition is a well established domain within affective computing and artificial intelligence. However, the categorical structures that underpin modern systems remain largely unchanged since the advent of universalist models and dimensional circumplex theories. Originating predominantly from a single cultural and epistemological tradition, these frameworks create a systemic blind

spot. Affective states culturally significant within the Indian context fall almost entirely outside the representational capacity of current computational systems.

The *Navarasa* framework, formalised in the *Natya Shastra* and philosophically deepened by Abhinavagupta in the *Abhinavabharati*, identifies nine foundational states: *Shringara* (love), *Hasya* (mirth), *Karuna* (sorrow), *Raudra* (fury), *Veer* (heroism), *Bhayanaka* (terror), *Bibhatsa* (disgust), *Adbhuta* (wonder), and *Shanta* (serenity). Each state is associated with specific physiognomic markers or *Sthayi Bhavas*, preserved through continuous pedagogical transmission across generations of classical performing artists. This unbroken lineage has produced practitioners whose expressive output carries the rigour of a structured, supervised training curriculum.

NavrasaNet seeks to bring this profound knowledge into the computational domain. We present a culturally grounded pipeline combining rigorous dataset curation with modern deep learning architectures. Specifically, we utilize an EfficientNet B2 backbone to effectively capture subtle facial micro expressions without the parameter bloat of traditional deep convolutional networks. Furthermore, we demonstrate the real world applicability of this emotion recognition system through the development of a **Vedic Raga Recommender Raga Ratnakar** : *Where Every Mood Finds a Raga* website, which utilizes the user’s inferred Navarasa state to suggest contextually resonant musical compositions.

2 The Navarasa Facial Expression Corpus

Existing benchmarks for facial affect recognition such as FER2013 or AffectNet are constructed exclusively around Western emotion categories and contain no culturally annotated samples representing Indian aesthetic traditions. To address this gap, we developed the Navarasa Facial Expression Corpus (NFEC) in partnership with performing arts institutions and individual practitioners whose credentials are recognized at the national level by the Government of India.

2.1 Curation and Authenticity

A foundational portion of the corpus was contributed by **Smt. Anita Babu**, an ICCR empanelled artist trained under Padmashree Guru Gangadhar Pradhan and Founder of Odissi Natya Sala, New Delhi. Her command of classical expressions brings immense credibility to the project ensuring that the emotions captured reflect the precision demanded by decades of rigorous training. Furthermore to ensure authenticity, the dataset’s expressive ground truth was verified by **Dr. Manorama B.N.**, Director, Researcher, and Author at Noopura Bhramari — an accredited Centre of Indian Knowledge System (IKS) under the IKS Division, Govt of India. Her extensive background in performing, teaching, researching and publishing multimedia literature in the realm of Indian Performing Arts establishes a rigorous academic and cultural validation layer. Additional contributions from practitioners across multiple classical dance forms (such as **Ms. Achuta Manasa**) provide cross tradition coverage and meaningful variation within each category.

2.2 Dataset Statistics and Split Strategy

The NFEC consists of exactly 21,752 carefully extracted and annotated frames. Traditional frame extraction via OpenCV frequently resulted in temporal corruption and early EOF errors. Thus, a custom byte level extraction pipeline utilizing Ffmpeg was engineered to guarantee the fidelity of the facial geometries.

To rigorously test the generalizability of the model against demographic biases, the dataset also utilised **Leave One Subject Out (LOSO)** cross validation paradigm. The data was split into Training (14,805 images), Validation (3,635 images), and Testing (3,312 images). The test split contains unseen performers, ensuring that the network learns the universal geometric structures of the *Navrasas* rather than merely memorizing the identities of the training subjects.

3 Computational Framework

Initial experiments on our custom dataset highlighted the immense difficulty of culturally specific emotion recognition. When building a standard convolutional network entirely from scratch, performance rapidly collapsed to random guessing (approximately 14% test accuracy). Conversely, transferring heavy architectures like ResNet 50 resulted in severe catastrophic overfitting: while the model achieved 98% accuracy on the validation set, it plummeted to a mere 60% on the test set, demonstrating a complete failure to generalize across different performers. To overcome these limitations and optimize the parameter to accuracy ratio for subtle facial feature extraction, NavrasaNet specifically employs an **EfficientNet B2** architecture.

3.1 Model Architecture

EfficientNet relies on compound scaling, uniformly scaling depth, width and resolution to capture fine grained facial geometries (such as the arching of an eyebrow in *Adbhuta* or the flaring of nostrils in *Raudra*) far more effectively than older architectures. The pre trained ImageNet classifier head was discarded and replaced with a densely connected custom head comprising dropout layers and a final 9 class Softmax layer corresponding to the Navarasa states.

3.2 Two Phase Training Strategy

Training deep networks on highly subjective emotional data frequently leads to catastrophic forgetting or early plateauing. To mitigate this, a two phase fine tuning strategy was implemented:

1. **Phase 1 (Warmup):** The EfficientNet B2 backbone is frozen and only the randomly initialized custom classification head is trained. This isolates the backbone’s pre learned texture detection capabilities from being destroyed by initially large, unstructured error gradients.

2. **Phase 2 (Full Fine Tuning):** The entire network is unfrozen. To ensure stable convergence, we employ a **Cosine Annealing Learning Rate Scheduler** paired with differential learning rates (backbone at 1×10^{-5} and classification head at 1×10^{-4}). The cosine decay smoothly anneals the learning rate, allowing the network to traverse the complex loss landscape efficiently without abrupt magnitude drops, ultimately settling into an optimal local minimum. Additionally, gradient clipping is applied to stabilize weight updates during the initial transition into full fine tuning.

3.3 Regularization for Subjective Emotion

Emotion is phenomenologically ambiguous; a practitioner portraying *Raudra* (Anger) may exhibit microscopic, overlapping traces of *Veer* (Heroism). To account for this inherent subjectivity, we employ **Label Smoothing** ($\alpha = 0.1$). This prevents the network from over calibrating to single class hard assignments, training the model to respect the continuous spectrum of affective states. Furthermore, aggressive spatial augmentations (Random Resized Crops, Color Jittering and Rotations) coupled with targeted dropout regularization ($p = 0.4$ and $p = 0.3$ in the dual layer classification head) enforce strict background and demographic invariance.

4 Experiments and Results

The model was trained on cloud based GPU instances with custom fault tolerance mechanisms to survive network interruptions. An automatic early stopping protocol halted training when validation accuracy plateaued, finalizing optimal model weights at Epoch 29.

4.1 Quantitative Results

NavrasaNet achieved a peak validation accuracy of **93.43%** and a robust test accuracy of **87.32%**. The stability of the training process, facilitated by the two phase fine tuning and cosine annealing learning rate scheduler is evident in the learning curves (Figure 1). Table 1 details the per class precision, recall and F1 scores on the test set.

4.2 Qualitative Analysis and Interpretability

Analysis of the confusion matrix (Figure 3) reveals strong diagonal separability, with robust discrimination for states such as *Shanta* and *Veera* that lack direct equivalents in Western taxonomies. Inter class confusions are largely biologically grounded; for instance, the minor confusion between *Raudra* and *Veer* correlates directly with shared facial muscle tension profiles (e.g. flared nostrils, intense gaze).

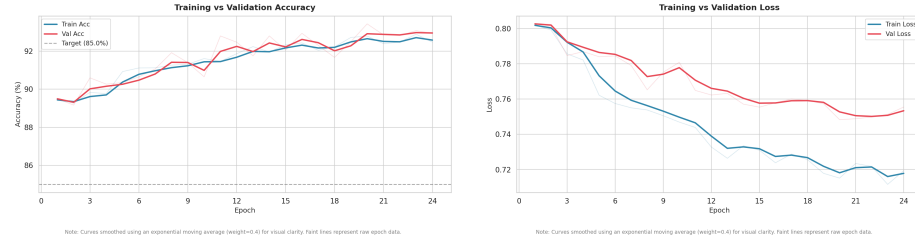


Fig. 1. Training vs validation curves of accuracy and loss demonstrates stable convergence without major overfitting.

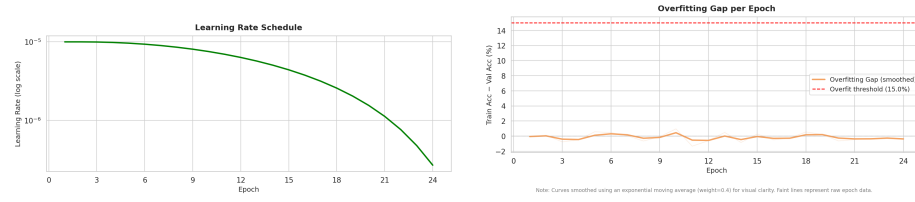


Fig. 2. Cosine Annealing learning rate schedule (left) and the resulting overfitting gap over epochs (right), validates the model's resistance towards memorizing training set.

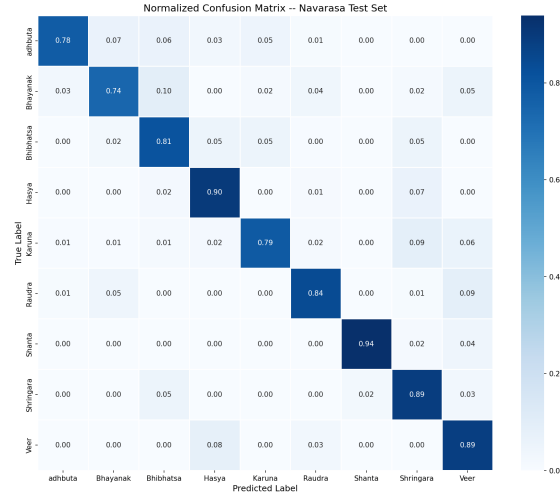
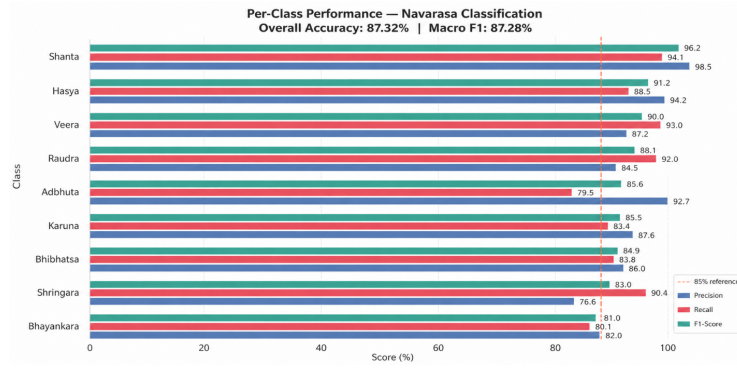


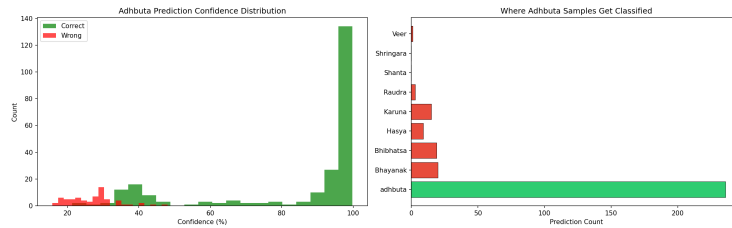
Fig. 3. Confusion matrix details the test set prediction performance across each Rasa.

Table 1. Classification Performance across the Navarasa categories on test set.

Rasa	Accuracy (%)	Precision	Recall	F1 Score
Shringara	90.42%	0.77	0.90	0.83
Hasya	88.46%	0.94	0.88	0.91
Karuna	83.43%	0.88	0.83	0.85
Raudra	92.02%	0.84	0.92	0.88
Veera	93.04%	0.87	0.93	0.90
Bhayanaka	80.10%	0.82	0.80	0.81
Bibhatsa	83.76%	0.86	0.84	0.85
Adbhuta	79.54%	0.93	0.80	0.86
Shanta	94.13%	0.98	0.94	0.96
Overall / Macro Avg.	87.32%	0.88	0.87	0.87

**Fig. 4.** Per class performance breakdown bar charts.

To provide deeper insight into class specific discriminative features, we extracted characteristic examples. Figure 5 demonstrates the model’s predictive breakdown for the *Adbhuta* (Wonder) class, confirming strong confidence allocation to the primary class.

**Fig. 5.** Detailed analysis illustrating the confidence distribution for *Adbhuta* samples.

To guarantee that the network relies on authentic facial expressions rather than background artifacts, Grad CAM visualisations were generated (Figure 6). The resulting heatmaps confirm that the EfficientNet B2 backbone actively anchors its predictions to eyes, eyebrows and mouth. Furthermore, it illustrates the model’s robust predictive capabilities in real world scenarios, effectively analyzing unseen frames to output confident, culturally accurate Navarasa classifications.



Fig. 6. Grad CAM heatmap analysis.

5 Application: The Vedic Raga Recommender

The ability to computationally detect these culturally specific states opens concrete avenues for human computer interaction. We have developed a full stack

web application, **Raag Ratnakar**, which directly leverages the NavrasaNet pipeline in real time. As demonstrated in Figure 9 and 10, by capturing the user’s current facial expression via a webcam interface, the system infers their prevailing Navarasa state and subsequently recommends a contextually appropriate musical composition (Raga) from the Indian classical tradition. This system effectively grounds the digital listening experience in ancient classical correspondences, offering a personalized, culturally responsive environment that dynamically adapts to the user’s affective state.

5.1 System Architecture

Raag Ratnakar is engineered as a modular, service oriented full stack web application. The backend exposes a RESTful API layer through which the following services are orchestrated:

1. **Rasa Predictor Service:** The NavrasaNet model is utilised to detect user’s current mood. Incoming webcam frame is pre processed (Resize to 256, CenterCrop to 224) and fed through the network to yield a Navarasa classification.
2. **Environment Service:** Computes the user’s temporal and seasonal context (detailed in Section 5.2).
3. **Mode Engine:** Implements the recommendation strategy (Traditional or Therapeutic, detailed in Section 5.3).
4. **Rasa Repository:** Maintains an indexed lookup of Raga to Sthayi Bhava mappings, enabling efficient retrieval of candidate Ragas by emotional affinity, seasonal association and temporal suitability.
5. **Raga Recommender:** Aggregates the outputs of the above services and applies a weighted scoring function (Section 5.4) to rank and return the final Raga recommendations.
6. **Authentication Service:** Manages user registration and session handling via JWT tokens with bcrypt hashed passwords, ensuring secure and persistent user profiles.

5.2 Uddipana: Environmental Emotional Triggers

A central innovation of Raag Ratnakar is its incorporation of *Uddipana* (excitants), i.e. environmental stimuli that amplify or modulate the listener’s emotional state. The *Natya Shastra* recognizes that aesthetic experience is not solely a function of an internal emotional disposition but is deeply conditioned by the surrounding environment. Two classical dimensions of Uddipana are computationally operationalized:

Prahara (Time of Day): Indian classical music prescribes specific performance times for each Raga, rooted in the concept of *Samay* (time). The system determines the user’s current *Prahara* by first obtaining the device’s GPS coordinates (latitude, longitude) and computing the exact local sunrise and sunset

times. The resulting daylight and nighttime durations are each divided into four equal segments, yielding the eight classical Praharas dynamically calibrated to the user’s geographic location rather than relying on fixed clock hours. This ensures that the temporal context remains astronomically accurate regardless of the user’s position on the globe.

Ritu (Season): To capture the seasonal dimension, the system queries the Open Meteo weather API using the user’s coordinates to retrieve live meteorological data encoded as WMO weather codes. These codes are mapped to qualitative weather conditions (e.g. WMO code 80 corresponds to a monsoon like rainfall pattern). Simultaneously, the current calendar month is converted to the corresponding Indian *Ritu* (e.g., *Varsha*, *Sharad*, *Hemanta*). A weather match score is then computed by cross referencing the API derived weather mood against the calendrical Ritu, capturing the degree of alignment between the actual atmospheric conditions and the classical seasonal ontology.

5.3 Recommendation Modes

Raag Ratnakar offers two philosophically distinct recommendation strategies, each grounded in classical Indian aesthetic theory:

Traditional Mode (ISO Principle + Sadharanikarana): This mode operates on the *ISO Principle* of music therapy, which advocates matching the listener’s current emotional state before gradually transitioning them. If the user is experiencing *Krodha* (anger), the system first recommends a Raga whose dominant Bhava resonates with anger, meeting the listener where they are emotionally. The philosophical justification derives from Abhinavagupta’s concept of *Sadharanikarana* (universalization of emotion): fully experiencing an emotion within an aesthetic frame is itself cathartic and healing, enabling a natural progression toward equanimity.

Therapeutic Mode: In contrast, the therapeutic strategy pivots immediately to a positive emotion at an equivalent energy level then gradually eases the listener toward a state of calm. This mode assigns equal weight (50/50) between energy matching and valence correction, prioritizing immediate mood elevation. A disclaimer accompanies this mode, clarifying that it is not a substitute for professional mental health support and is intended solely as an aesthetic wellness tool.

5.4 Scoring and Ranking

Once the Mode Engine determines the target emotion, candidate Ragas from the repository are ranked using a composite weighted score:

- **Bhava Match (70%):** Quantifies the alignment between the Raga’s primary or secondary *Sthayi Bhava* and the target emotion generated by the Mode Engine.
- **Ritu & Weather Match (20%):** Evaluates the correspondence between prevailing meteorological conditions and the Raga’s traditional seasonal associations.
- **Prahara Match (10%):** Verifies that the user’s local time of day falls accurately within the Raga’s prescribed performance window.

This weighting ensures that emotional resonance remains the dominant selection criterion while environmental and temporal context provide meaningful secondary modulation.

The application additionally provides pre built activity modes for specific use cases such as Gym, Study, Sleep, and Meditation, each mapping to a curated set of Ragas selected on the basis of peer reviewed neuroscience research (Figures 7 and 8).

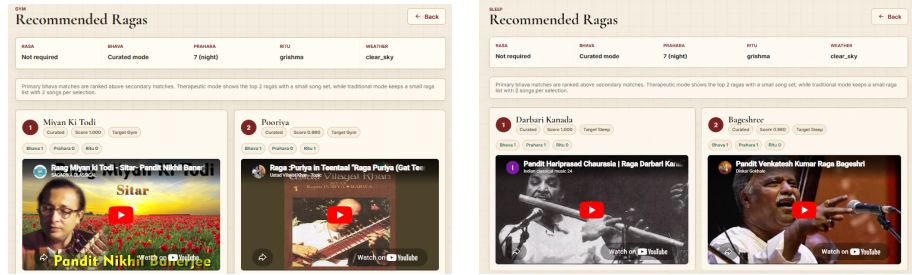


Fig. 7. Activity mode interfaces: Gym Mode (left) and Sleep Mode (right).

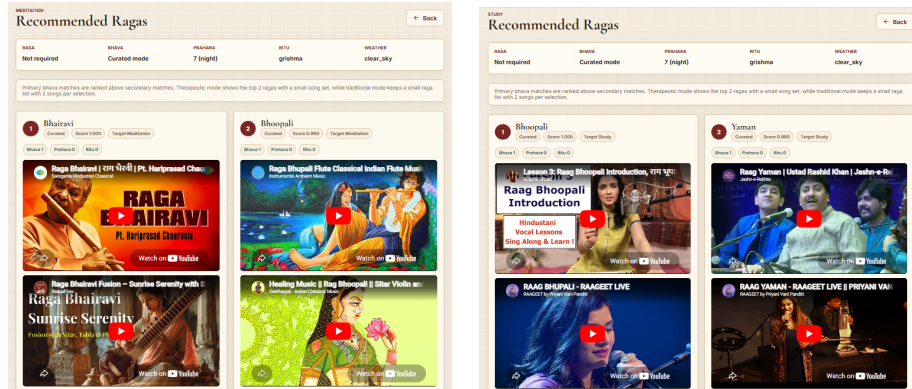


Fig. 8. Activity mode interfaces: Meditation Mode (left) and Study Mode (right).

SWARASTHANA

Raga Recommender

Email

Password

→] Sign in

New here? **Create account**

SAMAYA AND BHAVA

Prepare recommendation



Therapeutic

Guides the recommendation toward steadier emotional balance.



Traditional

Follows a classical rasa journey while preserving the detected mood.



Study

Curated ragas for focus and sustained attention.



Gym

Rhythmic and energetic ragas for movement.



Sleep

Late, calm ragas for rest and slowing down.



Meditation

Gentle ragas for stillness and internal quiet.



THERAPEUTIC

Recommended Ragas

← Back

⚠ Detected rasa is uncertain. The list uses the closest bhava match.

RASA	BHAVA	PRAHARA	RITU	WEATHER
shringara	rati	8 (night)	grishma	clear_sky

Primary bhava matches are ranked above secondary matches. Therapeutic mode shows the top 2 ragas with a small song set, while traditional mode keeps a small raga list with 2 songs per selection.

1 Ahir Bhairav

Primary Score 0.850 Target Rati

Bhava 1 Prahara 1 Ritu 0

Raga Ahir Bhairav | Mahesh Kale | Indian Classical

Raga Ahir Bhairav

Mahesh Kale

Watch on YouTube

Ahir Bhairav - Preview | Kaushiki Chakraborty |

Raga Ahir Bhairav

KAUSHIKI CHAKRABORTY

Watch on YouTube

2 Bhimpalasi

Primary Score 0.800 Target Rati

Bhava 1 Prahara 0 Ritu 1

Bhimpalasi | Anirudh Varma Collective | Aanchal

Bhimpalasi

Watch on YouTube

Rashid is Rashid Vol. 1 Raag Bhimpalasi

Rashid is Rashid

RAAG BHIMPALASI

Watch on YouTube

Fig. 10. Personalised Raga Recommendations based on users's current mood, prahara timing, location and weather.

6 Conclusion

NavrasaNet presents a computationally grounded approach to culturally specific emotion recognition, supported by a practitioner validated corpus whose annotation quality derives from expert consensus within living classical traditions. By engineering a custom data pipeline, leveraging the compound scaling of EfficientNet B2 and utilising LOSO cross validation, we achieved 87.32% recognition accuracy across nine subjective states. Future work aims to expand the framework into a multimodal system incorporating vocal prosody and gestural cues, reflecting the holistic nature of Indian aesthetics.

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